L#04: Kirchhoff's laws

2018-02-12

Kirchhoff's current law (KCL):

algebraic sum of currents entering or leaving a node or (a closed boundary) is $\boxed{\text{zero}}$.

Mathematical description:

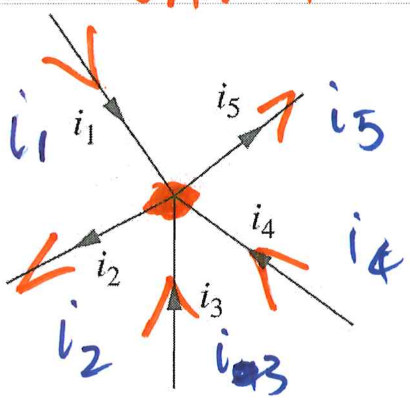
$$\sum_{n=1}^{N} i_n = 0 ,$$

i_n: current in nth branch.

N: number of currents.

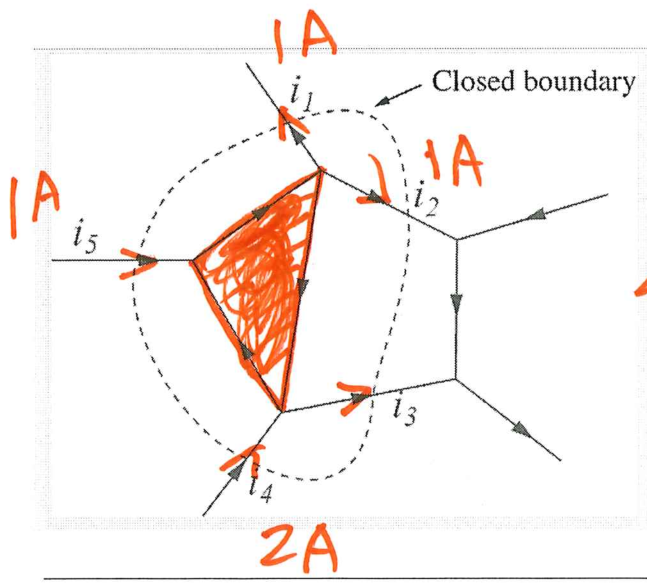
Sign Convention:

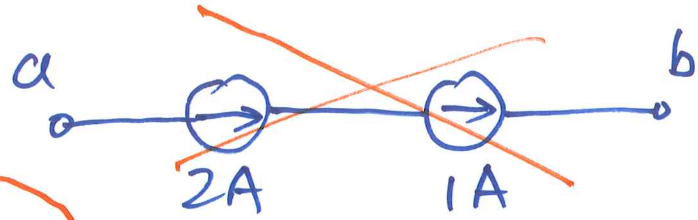
in: + ; out: −



$$i_1 - i_2 + i_3 + i_4 - i_5 = 0$$

$$\underbrace{i_1 + i_3 + i_4}_{\text{in}} = \underbrace{i_2 + i_5}_{\text{out}}$$

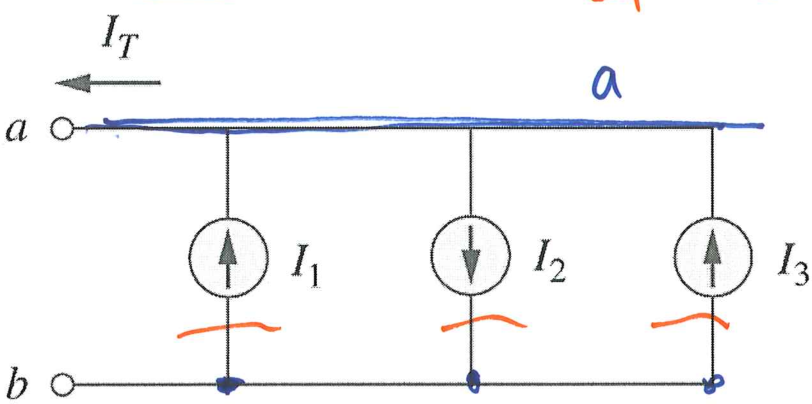




Current sources in parallel

$I_1 \neq I_2 \neq I_3$

1



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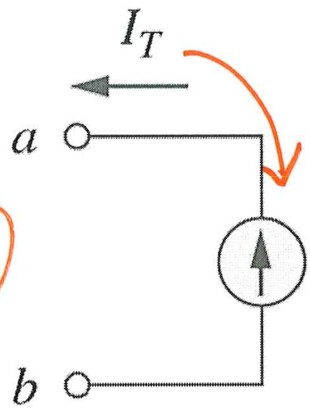
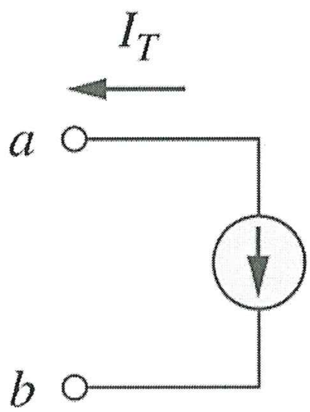


$I_T^1 = I_1 - I_2 + I_3$

I_T^1

$I_T^1 = -I_T^2$

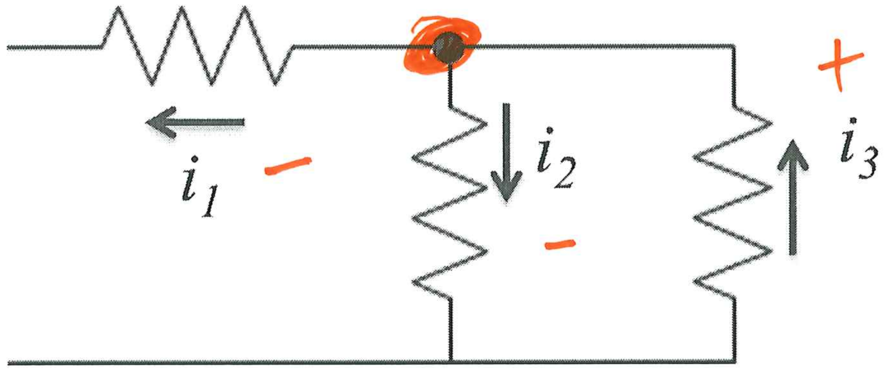


I_T^2

$I_T^2 = -I_1 + I_2 - I_3.$

$= -(I_1 - I_2 + I_3)$

KCL: resistive networks

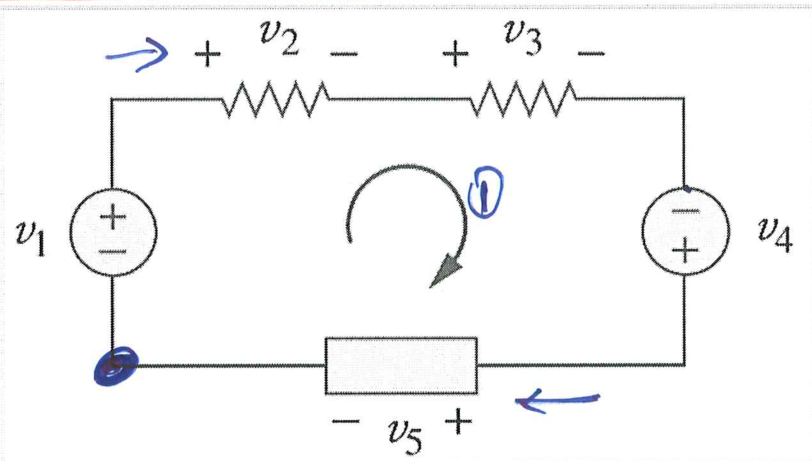


$$-i_1 - i_2 + i_3 = 0. \quad (1)$$

$$i_3 = i_1 + i_2 \quad (2)$$

Kirchhoff's voltage law (KVL): algebraic sum of voltages around a closed loop is 0.

$$\boxed{-V_1 + V_2 + V_3 - V_4 + V_5 = 0}$$



Sign Convention: ① we let current flows clockwisely; ② current enters "+" polarity; ③ when meet "−", note "−", when meet "+", note "+".

Mathematical Description:

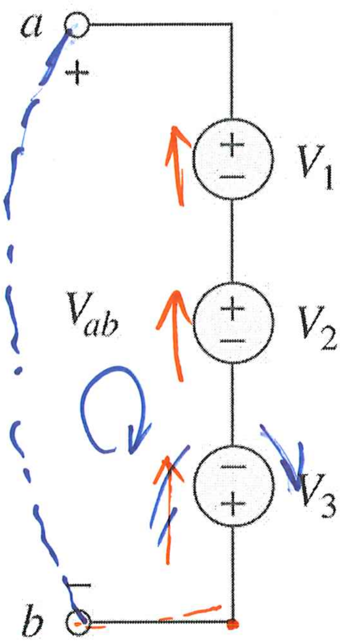
$$\sum_{i=1}^{N} V_i = 0$$

V_i: voltage across ith element.

N: number of elements.

Sign convention applies!

Voltage source in series

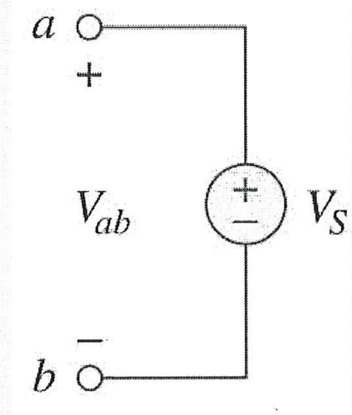


① $-V_{ab}+V_1+V_2-V_3=0$.

$V_{ab}=V_1+V_2-V_3$

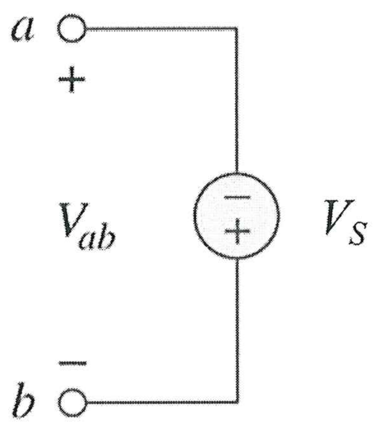
② $V_{ab}=V_a-V_b$, $V_b=0$.

$V_a=-V_3+V_2+V_1$



$-V_{ab}+V_s=0$

$\Rightarrow V_{ab}=V_s$

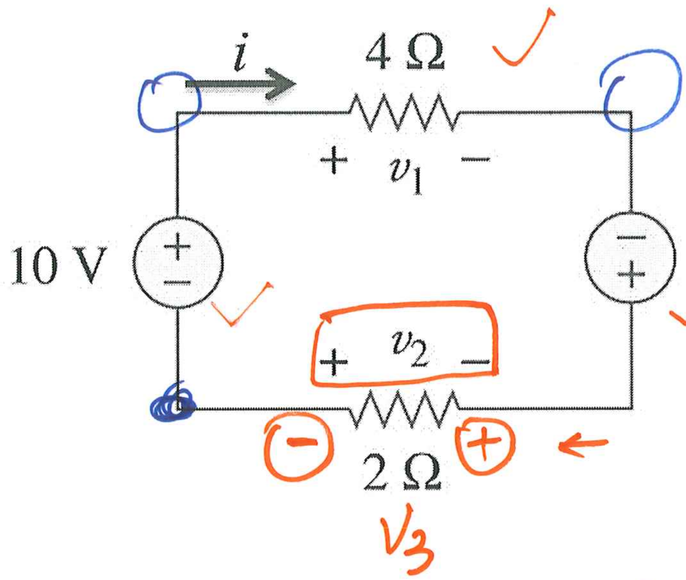


$-V_{ab}-V_s=0$

$\Rightarrow V_s=-V_{ab}$.

Example #1:

Find i, v_1 and v_2.



$V_2 = -V_3$.

KVL: $-10 + V_1 - 8 + V_3 = 0$

$V_1 + V_3 = 18$

$4i + 2i = 18 \Rightarrow i = 3A$

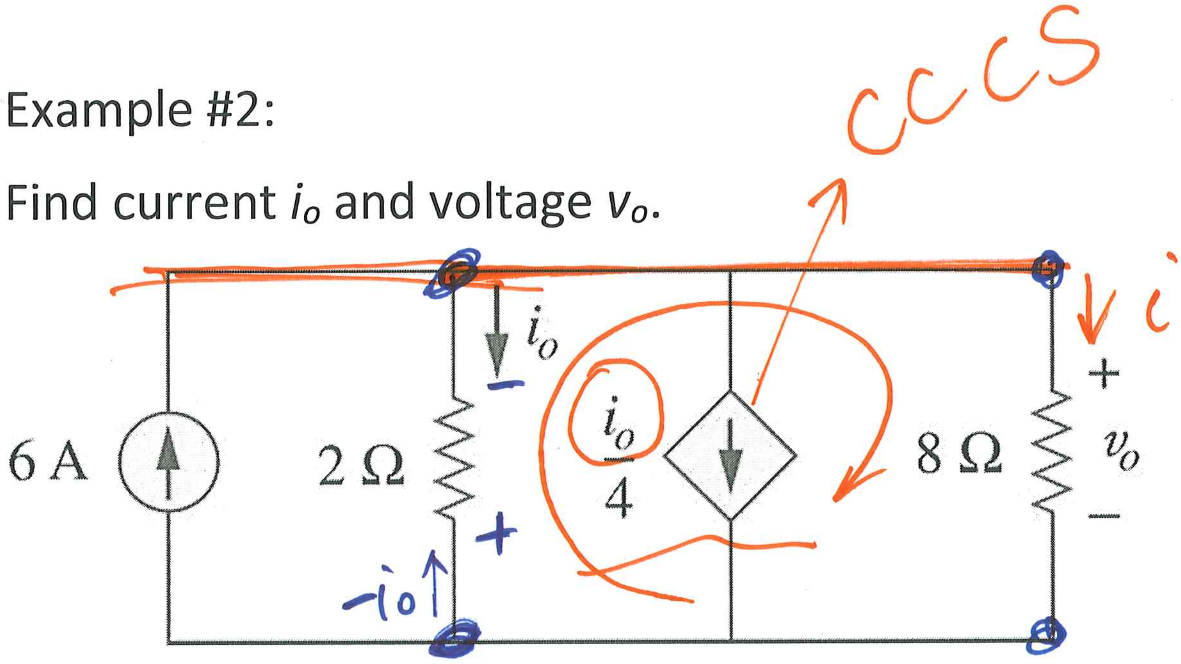
$V_1 = 4i = 12V$.

$V_3 = 2i = 6V$.

$V_2 = -V_3 = -6V$.

Example #2:

Find current i_o and voltage v_o.



$$6A - i_o - \frac{i_o}{4} - \frac{V_o}{8} = 0$$

$$-2i_o + V_o = 0 \Rightarrow \underline{V_o = 2i_o}$$

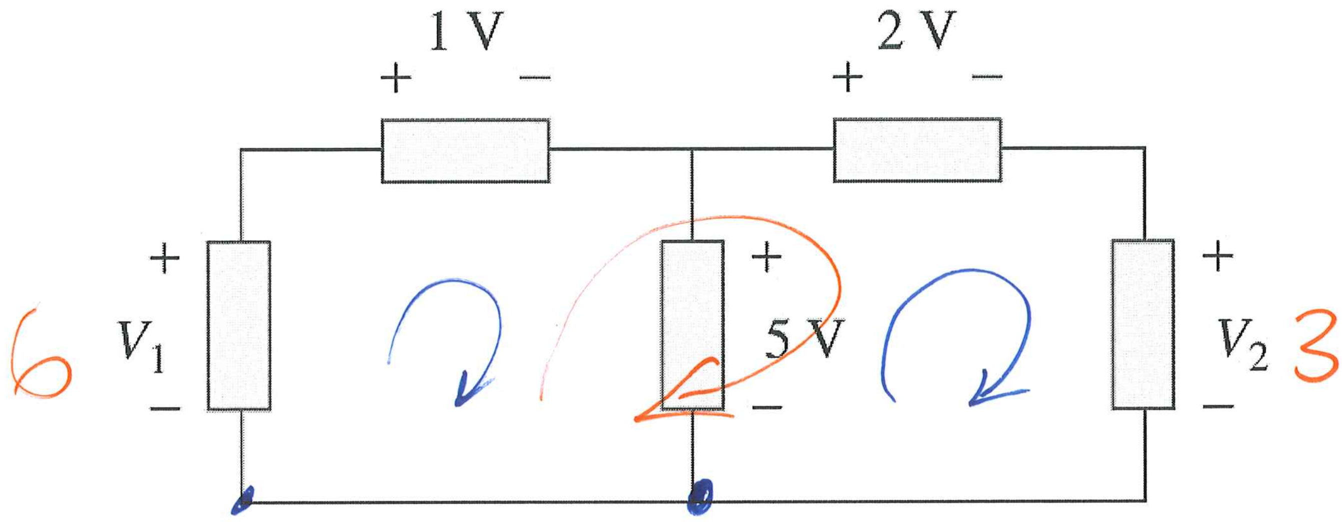
$$6 - i_o - \frac{i_o}{4} - \frac{i_o}{4} = 0 \Rightarrow$$

$$1.5\,i_o = 6 \quad \Rightarrow i_o = 4A$$

$$V_o = 8V.$$

Example #3:

Calculate V_1 and V_2.



$-V_1 + 1 + 5 = 0 \Rightarrow V_1 = 6\,V.$

$-5 + 2 + V_2 = 0 \Rightarrow V_2 = 3\,V.$

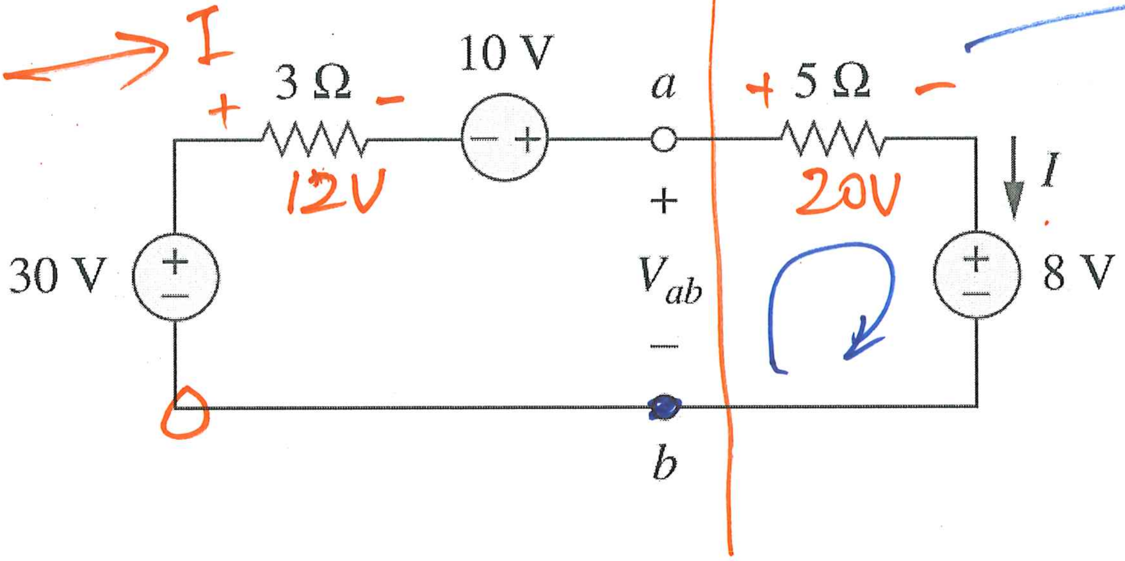
$\underline{-6 + 1 + 2 + 3} = 0$ ✓

Example #4:

Determine I and V_{ab}.

$$-V_{ab}+20+8=0$$

$$V_{ab}=28V.$$



$$-30+\underline{3I}-10+\underline{5I}+8=0$$

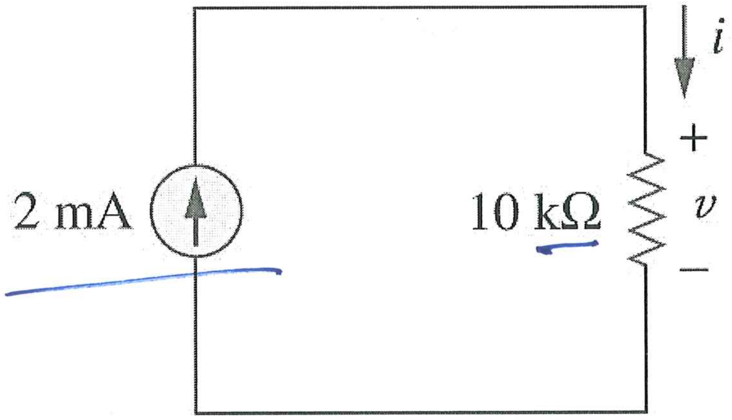
$$8I=32 \Rightarrow \underline{I=4A}$$

$$-30+12-10+V_{ab}=0$$

$$\Rightarrow \underline{V_{ab}=28V.}$$

Example #1:

For the circuit shown, calculate the voltage *v*, conductance *G* and power *p*.



$$V = iR = 2 \times \cancel{10^{-3}}\,\underline{A} \times 10 \times \cancel{10^{3}}\,\underline{\Omega} = 20\ V$$

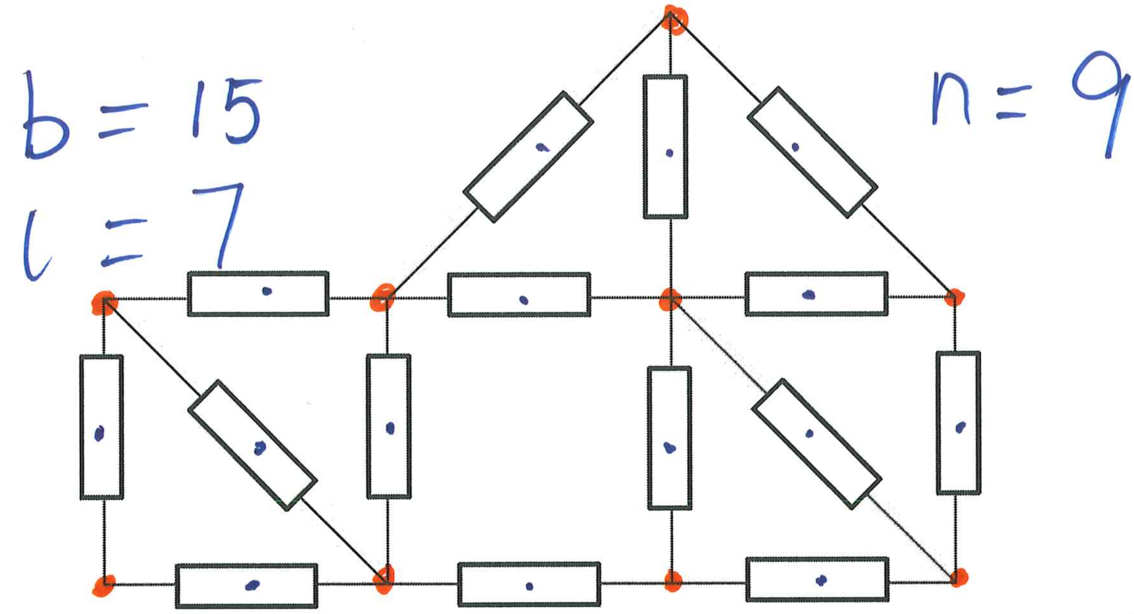
$$G = \frac{1}{R} = \frac{1}{10 \times 10^{+3}} = 0.1\ mS.$$

$$P = i^2 R = v \cdot i = 20V \cdot 2mA$$

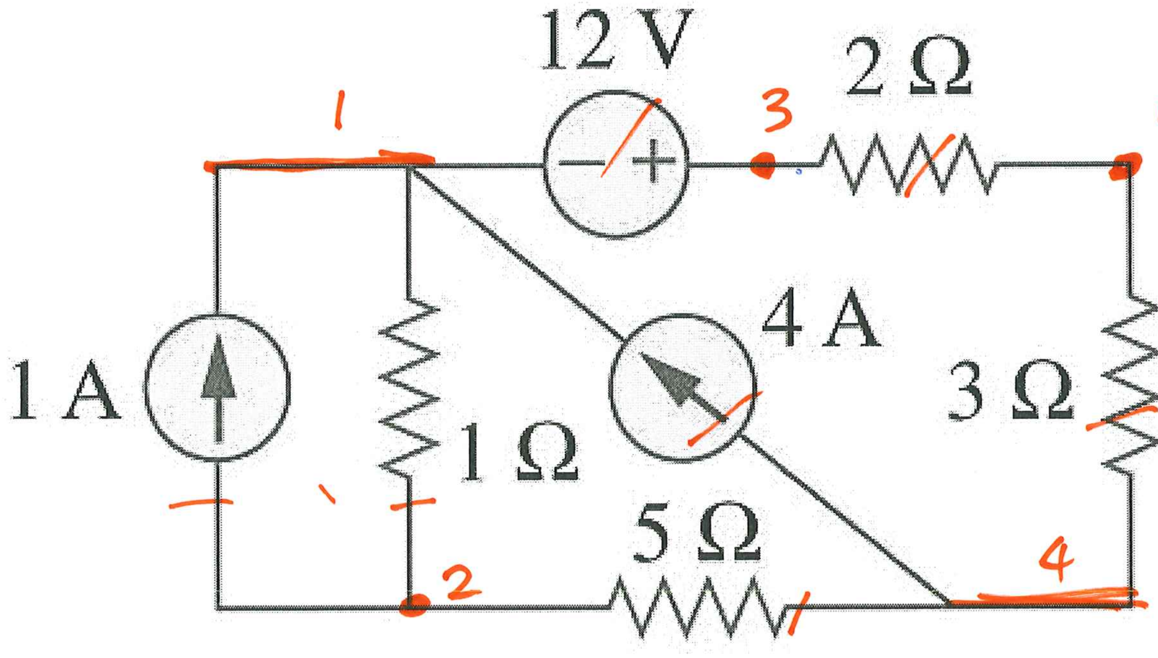
$$= 40\ mW.$$

Example #2:

Find the number of nodes, branches and independent loops.



$b = l + n - 1 \Rightarrow \underline{15 = 7 + 9 - 1}$ ✓



$b = 7,\ n = 5,\ l = 3$ ✓